

universität freiburg



7TH ENZYMEML WORKSHOP · RÜDESHEIM · HANDS-ON SESSION 2 · 16 SEPTEMBER 2026

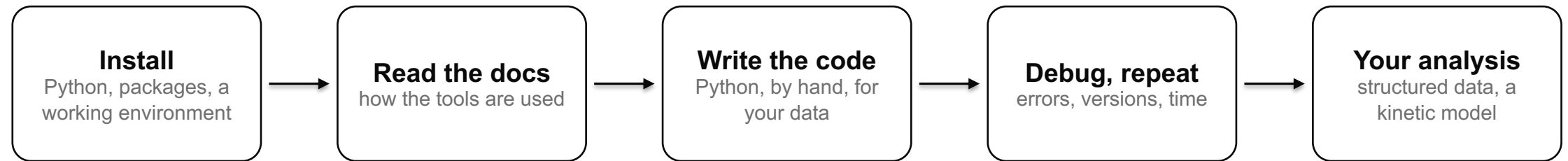
# From measurement to kinetic modelling

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# Last year the code was the hurdle

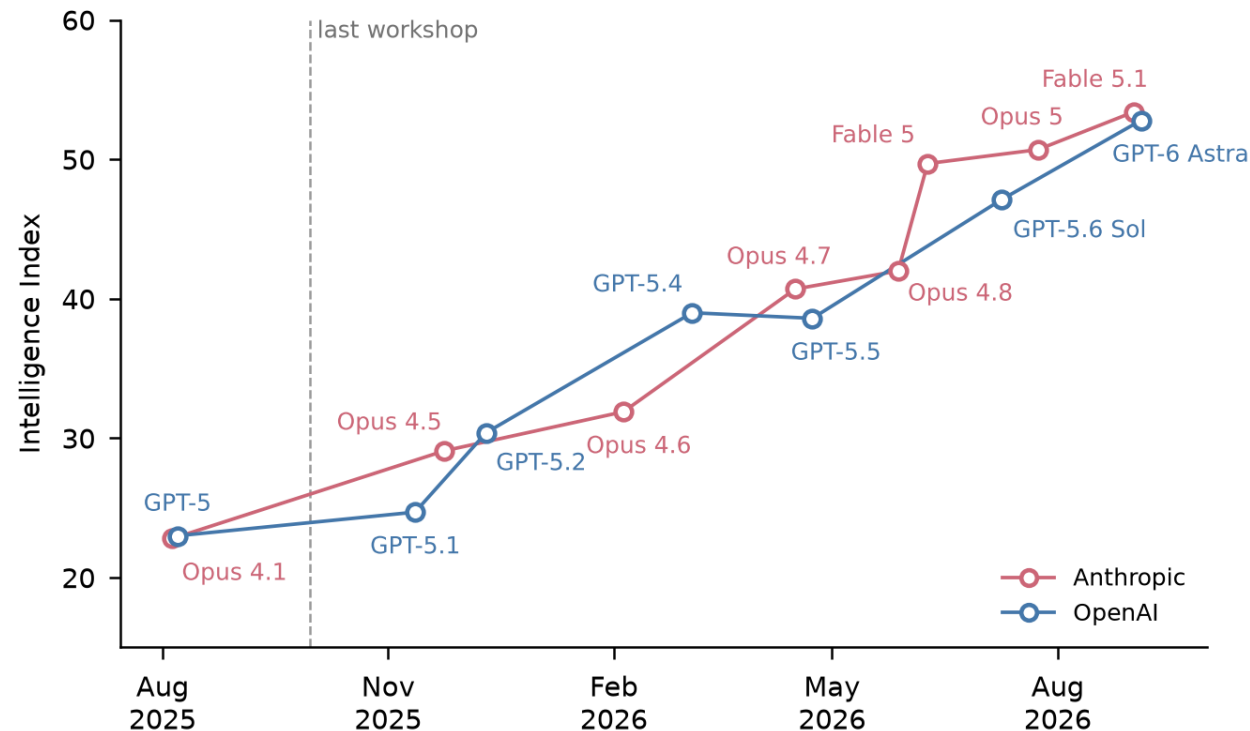
Previous hands-on sessions: we installed, read the docs and wrote the analysis code ourselves



- Failing installs and unfamiliar code: a hurdle for everyone who does not code daily
- So the data stayed unstructured, and the contemporary methods out of reach
- **Since then:** the models got better, and so did the harnesses that let them run tools

# From 23 to 53 in one year

The two labs traded the lead several times; the biggest single step was Fable 5 in June 2026



- The Intelligence Index is an independent score from 0 to 100 that combines many tests of knowledge, reasoning, coding and agentic work, with every model tested the same way.
- Shown are the flagship models of Anthropic and OpenAI that anyone can use, each at its strongest public setting.

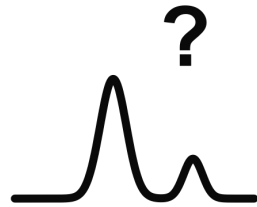
Artificial Analysis Intelligence Index v4.3, artificialanalysis.ai; release dates from the vendors

# The assistant types, you decide



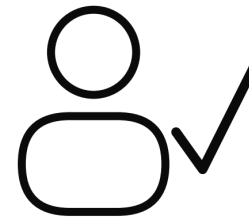
## One page is enough

From a one-page card, today's models write working code for our tools



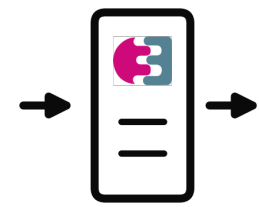
## Blind to your data

Where the peak is, which model is right, when a result is wrong



## You decide and check

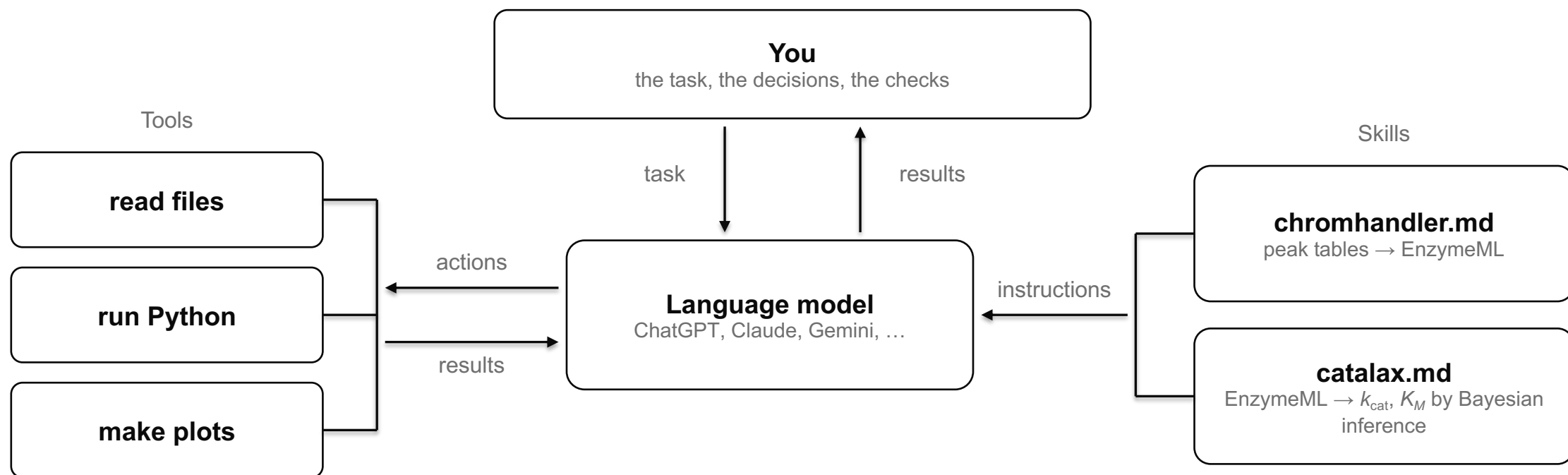
The decisions and the checks stay with the scientist; the assistant executes



## EnzymeML in between

One document carries the data between the steps and the tools

# Tools act, skills instruct, you decide

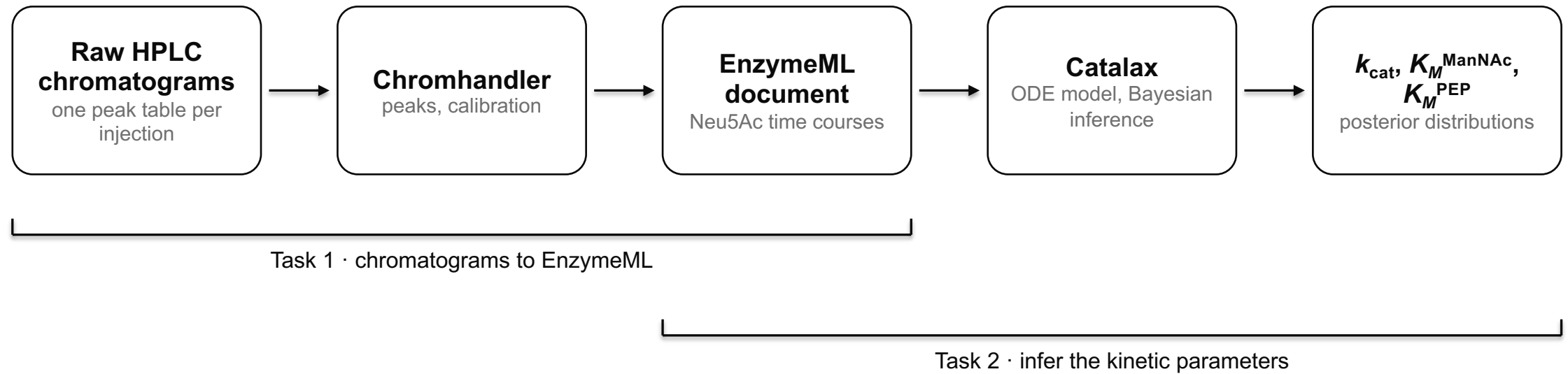


- Tools let the model read the data and run code; skills tell it how our libraries turn data into insight

## WORKFLOW

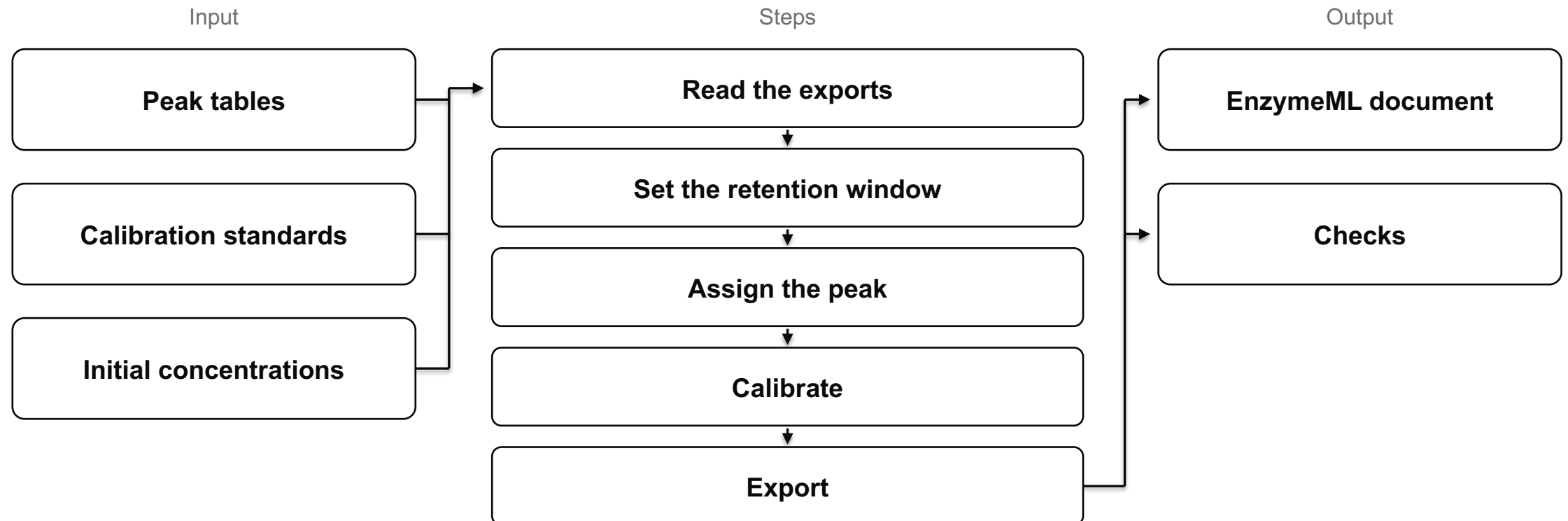
# Two tasks, one document in between

Task 1 ends with the EnzymeML document; task 2 starts from it



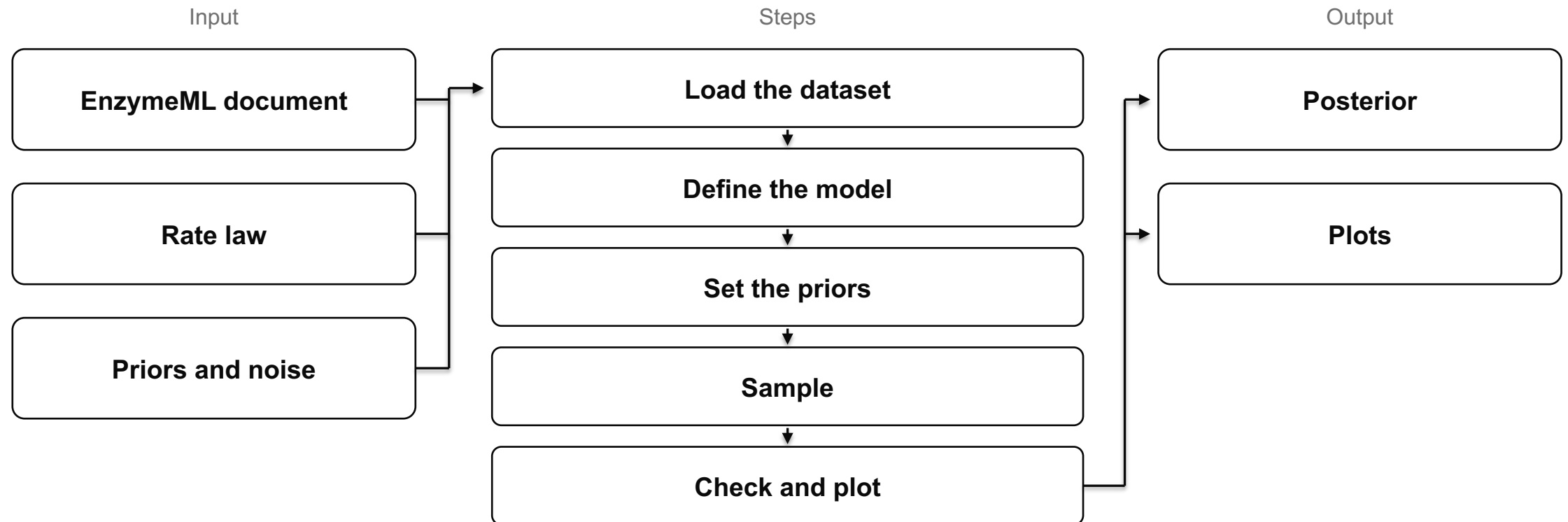
# Chromhandler: from peak tables to concentrations

Reads the instrument's exports, assigns and calibrates one peak per compound, writes EnzymeML



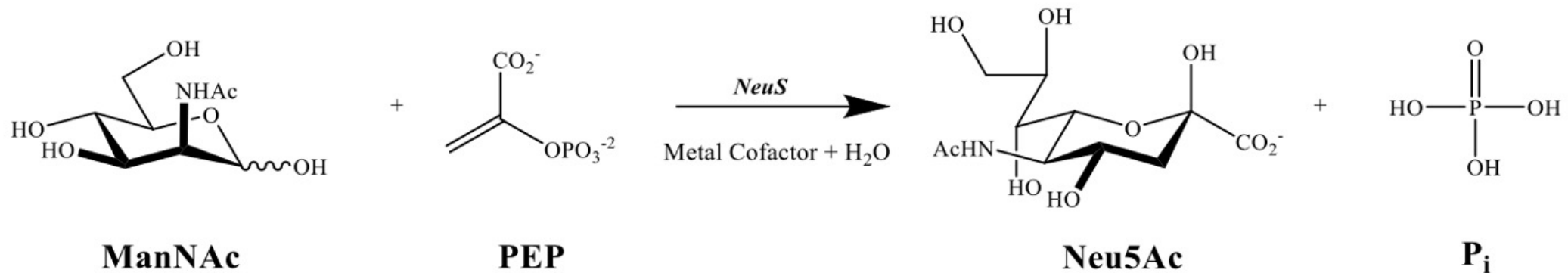
# Catalax: from time courses to kinetic parameters

Builds the ODE model from a rate law and samples the posterior of its parameters



## THE EXPERIMENT

# ManNAc + PEP → Neu5Ac + Pi, followed by HPLC



- Reactions sampled over time and followed by HPLC-PDA at 215 nm
- What you get is the instrument's peak table per injection, not the raw trace
- Only the product Neu5Ac is quantified; calibration standards with known concentrations
- How many reactions, which conditions, how they were varied: that is in the data

Data: Çakar et al. (2024) Kinetic characterization of two neuraminic acid synthases and evaluation of their application potential. Appl. Microbiol. Biotechnol. 108, 446 · doi:10.1007/s00253-024-13277-1

TODAY

# What we do today

- **Task 1:** from the peak tables to an EnzymeML document (Chromhandler)
- **Task 2:** from the EnzymeML document to  $k_{\text{cat}}$  and  $K_M$  by Bayesian inference (Catalax)
- A live demo of each task first, then you work in pairs
- The checks in the task card say when you are done
- At the end we compare everybody's parameters and discuss

# One repository, two ways to work

[tinyurl.com/enzymeml2026](https://tinyurl.com/enzymeml2026) · start with the README

## With a language model

- **Any chat model**, free or by subscription: ChatGPT, Claude, Gemini, ...
- **Paste** the tool card and the task card; run the code, paste errors back
- **The checks** decide when you are done
- **notebooks/** are the reference solutions, checkpoints/ the result of task 1

## Without writing code

- **Open the Colab link**: a prepared notebook, tools installed, data loaded
- **Run it cell by cell**; change a window, a prior, a series
- **Ask the model in Colab** to explain any cell

## TASK 1

# Task 1: from chromatograms to EnzymeML

Result: an EnzymeML document with the Neu5Ac time course of every reaction

## Tasks

- Plot the data
- Check how the kinetic assay was designed
- Calibrate: the calibration parameters
- Assign the product peak in the reactions
- Export the EnzymeML document
- **Details:** README, section Task 1

## How to do it

- **Coding assistant on your computer:** the cards in briefs/
- **Chat assistant only:** the Colab starter notebook
- **No assistant that runs code?** team up
- **Details:** README, section How to do it
- **Repository:** [tinyurl.com/enzymeml2026](https://tinyurl.com/enzymeml2026)

## TASK 2

# Task 2: kinetic parameters by Bayesian inference

$$v = k_{\text{cat}} \cdot [\text{NeuS}] \cdot [\text{ManNAc}] \cdot [\text{PEP}] / ((K_M^{\text{ManNAc}} + [\text{ManNAc}]) (K_M^{\text{PEP}} + [\text{PEP}]))$$

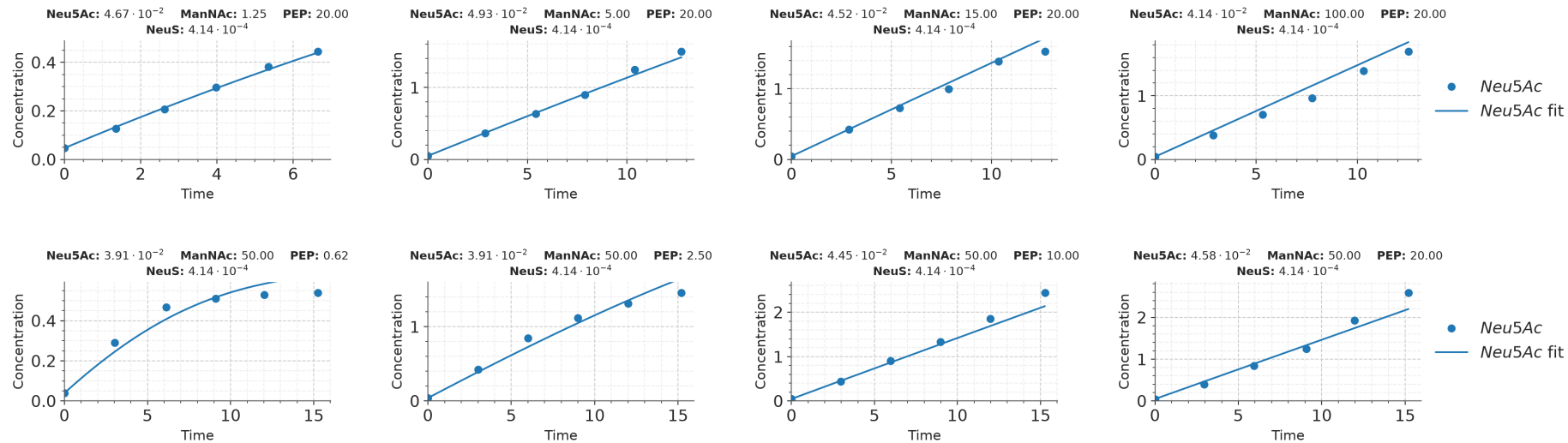
## Tasks

- Plot the time courses
- Define the model and the priors
- Infer the kinetic parameters and their uncertainty
- The correlation between the kinetic parameters
- Does one model describe both series?
- **Details:** README, section Task 2

## How to do it

- **Coding assistant on your computer:** the cards in briefs/
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# The reference fit, and an open question



| Parameter             | Posterior mean $\pm$ sd                                                    |
|-----------------------|----------------------------------------------------------------------------|
| $k_{\text{cat}}$      | $364 \pm 6 \text{ min}^{-1}$ ( $6.1 \text{ s}^{-1}$ , $9.4 \text{ U/mg}$ ) |
| $K_M^{\text{ManNAc}}$ | $1.53 \pm 0.19 \text{ mM}$                                                 |
| $K_M^{\text{PEP}}$    | $0.59 \pm 0.07 \text{ mM}$                                                 |

- One shared  $k_{\text{cat}}$  over-predicts the ManNAc series and under-predicts the PEP series
- The two series were measured on different days and differ by roughly 25 % in activity
- What would you do next?

# What to take home

- A one-page card is enough for an assistant to drive our tools
- The decisions and the checks stay with the scientist
- The cards are reusable as SKILL.md or AGENTS.md in coding agents
- What did not work for you today is the most useful feedback for us

# Thank you

[tinyurl.com/enzymeml2026](https://tinyurl.com/enzymeml2026)

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